Approach and Technologies Used

1. Objective

The objective of this project was to simulate an insurance dataset containing policy sales and claims data.

2. Technologies Used

Python was used as the primary programming language to generate and analyze the dataset.

Pandas was used for creating data tables, data manipulation, and analytical queries.

NumPy was used for generating large arrays, implementing tenure distributions, and randomly selecting claim records.

The final datasets are stored in CSV format for easy sharing and further analysis.

3. Policy Sales Data Generation

A synthetic dataset of 1,000,000 policyholders was generated.

Policies are sold equally across all days between January 1, 2024, and December 31, 2024.

Policy tenures are as per the required distribution.

1 Year - 20%

2 Years - 30%

3 Years - 40%

4 Years - 10%

Premium was calculated using the formula:

Premium = 100 * Policy Tenure

Policy start dates are set to 365 days after purchase, and policy end dates are calculated based on tenure.

4. Claims Data Generation

For the year 2025, it was assumed that 30% of the vehicles purchased on the 7th, 14th, 21st, and 28th of each month made claims on the start date of their policy.

For the year 2026, it was assumed that 10% of the vehicles with 4-year tenure policies made claims between January 1 and February 28, 2026.

The amount of the claim was assumed to be 10% of the total vehicle value.

5. Analysis

Analytical queries were conducted to determine the total premium, claim costs, and loss ratios, as well as the claim liabilities.